

Milestone 1 Report

CMPT310 - D200

Group 20

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Project Summary

The goal of this project is to develop an AI model capable of identifying and categorizing 50 dog breeds to help users make more informed decisions for their pets' health. We will use a Convolutional Neural Network which takes JPG image inputs and outputs the 3 most likely breeds for the given dog.

What Has Been Accomplished So Far

For Milestone 1, we created a simpler model which relies on logistic regression to classify a smaller sample size of 3 breeds. The overall accuracy obtained was 0.578, refer to Appendix A for full results.

Updates to the Data set

We are using the same dataset declared in the project proposal, but we narrowed it down to 3 distinct dog breeds for this milestone: Italian Greyhound, Mexican Hairless and Pomeranian.

We cut down the sample size for each breed to 150 images, for a total of 450 images. We used 90 for testing and 360 for training (80/20 split).

Preprocessing

To preprocess the images, we first resized and cropped them to fit into a smaller array of 64x64 pixels. We then turned them into grayscale, and normalized their pixel values on the range [0..1]. Finally, we flattened them into a vector. We also used PCA with 300 components with a C of 10.0 (because of the small dataset) and 2000 iterations..

Baseline model

We trained a simple logistic regression model on the 3 chosen dogbreeds, which we can use as a baseline model to benchmark and improve our final CNN model.

Additionally, we began planning which filters to use in our CNN layers, and researching how to implement them.

What Has Fallen Behind

Tasks or goals we expected to complete but didn't.

- We initially planned to set up the filters for the CNN model, but this proved to be highly scope too quickly before we had a feel for working with the data.

Challenges or blockers you've faced.

- We had to iterate through many dog breeds to find 3 that are distinct enough to make a reliable benchmark.
- We initially implemented the baseline model using KNN, but its performance was inconsistent, so we settled for a logistic regression model instead.

Project Changes

- Changes to the minimum viable system (milestone 1):
 - Swapped the AI technique to logistic regression
 - This change was done to reduce the scope of the minimum viable system and have a working model that can be used as a comparison point for future models.
 - Changed the goal from recognizing whether or not an image contained a dog to classifying 3 dog breeds.
 - The goal of the minimum viable system was changed to better match the core goal of our project. It is now a toy version of the main goal rather than a completely different problem.

Appendix A - Test results of the Baseline Model

This output was generated with the help of the Scikit-learn library methods.

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ACCURACY: 0.578 BASED ON THE 52/90 samples
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TECHNICAL REPORT USING SKLEARN CLASSIFICATION REPORT
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	precision	recall	f1-score	support
Italian_greyhound	0.47	0.27	0.34	30
Mexican_hairless	0.66	0.77	0.71	30
Pomeranian	0.55	0.70	0.62	30
accuracy			0.58	90
macro avg	0.56	0.58	0.56	90
weighted avg	0.56	0.58	0.56	90